Optimal Reconfiguration of Systems in Aeronautics

Lecture 07: "Newton-Kleinman Algorithm for the Algebraic Riccati Equation"

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Outline

1. Iterative Methods for Fault Accommodation
2. Initial Step Problem for the Newton-Kleinman Method
3. B747 Application Example

Objective I

This lecture provides some results on system stability.

It is mainly used when the Kleinman initialization method for obtaining the stabilizing initial feedback for the Progressive Accommodation Scheme is presented.

It introduces the system stability theorems in the sense of Lyapunov, while it also gives the stabilizing initial guess used in the Newton-Kleinman algorithm to solving the post-fault Continuous Algebraic Riccati Equation.

Outline

1. Iterative Methods for Fault Accommodation
   - Preliminaries on Lyapunov Stability
   - Preliminaries on Newton Method
   - Newton-Kleinman Approach
2. Initial Step Problem for the Newton-Kleinman Method
3. B747 Application Example

Lyapunov stability theorems I

Theorem (Gantmacher; 1959)

If all eigenvalues of matrix $A_f \in \mathbb{R}^{n \times n}$ have negative real-parts, then, to an arbitrary positive-definite quadratic form $V_f[x(t)]$, with $x(t) \in \mathbb{R}^n$, there corresponds a negative-definite quadratic form $W_f[x(t)]$, such that, if

$$\dot{x}(t) = A_f x(t) \,, \tag{1}$$

then $V_f[x(t)]$ and $W_f[x(t)]$ satisfy

$$\dot{V}_f[x(t)] = W_f[x(t)] \,. \tag{2}$$

Conversely, if, for some negative-definite form $W_f[x(t)]$, there exists a positive-definite form $V_f[x(t)]$, by (1) and (2), then all eigenvalues of matrix A_f have negative real-parts. □

Lyapunov stability theorems II

Theorem (Gantmacher; 1959)

In order that all eigenvalues of matrix $A_f \in \mathbb{R}^{n \times n}$ *have negative real-parts, it is necessary and sufficient that the matrix equation*

$$A_f^\top V_f + V_f A_f + Q_f = 0 \tag{3}$$

has as a solution the matrix of coefficients of some positive-definite quadratic form $V_f[x(t)] > 0$, *with* Q_f *any positive-definite matrix.* $\square$

Preliminaries on Newton method I

This part provides further results on the Newton-Kleinman algorithm.

It is useful when the Kleinman initialization method for obtaining the stabilizing initial feedback for the Progressive Accommodation Scheme is investigated.

It presents several supplementary conditions on the initial stabilizing gain for the Newton-Kleinman algorithm, while it also formulates the classical Newton-Raphson method, as well.

Kenney convergence theorems I

Lemma (Kenney, Laub, and Wette; 1990)

Let $X_{f_{N_{NK}}}$ *satisfy* $CARE_f$, $\hat{A}_f = A_f - B_f B_f^\top X_{f_{N_{NK}}}$ *is stable, and assume that* X_{f_0} *is near enough* $X_{f_{N_{NK}}}$, *such that*

$$2\|B_f\|_2^2 \left\| X_{f_{N_{NK}}} - X_{f_0} \right\|_2 \left\| \hat{\Omega}_f^{-1} \right\|_2 < 1 \,.$$

Then $\hat{A}_{f_0} = A_f - B_f B_f^\top X_{f_0}$ *is stable and*

$$\left\| \hat{\Omega}_f^{-1} - \hat{\Omega}_{f_0}^{-1} \right\|_2 \leq 2\|B_f\|_2^2 \left\| X_{f_{N_{NK}}} - X_{f_0} \right\|_2 \left\| \hat{\Omega}_f^{-1} \right\|_2 \left\| \hat{\Omega}_{f_0}^{-1} \right\|_2 ;$$

Kenney convergence theorems II

Lemma (Kenney, Laub, and Wette; 1990 – cont'd)

consequently

$$\frac{\left\|\hat{\Omega}_f^{-1}\right\|_2}{1+2\|B_f\|_2^2\left\|X_{f_{N_{NK}}}-X_{f_0}\right\|_2\left\|\hat{\Omega}_f^{-1}\right\|_2} \leq \left\|\hat{\Omega}_{f_0}^{-1}\right\|_2 \leq \frac{\left\|\hat{\Omega}_f^{-1}\right\|_2}{1-2\|B_f\|_2^2\left\|X_{f_{N_{NK}}}-X_{f_0}\right\|_2\left\|\hat{\Omega}_f^{-1}\right\|_2},$$

where $\left\|\hat{\Omega}_{f_0}^{-1}\right\|_2$ *stands for the operator-norm of the inverse closed-loop Lyapunov-operator of the initial-guess for the Newton-Kleinman approach, that is*

$$\hat{\Omega}_{f_0}(Y_{f_0}) \triangleq (A_f - B_f B_f^\top X_{f_0})^\top Y_{f_0} + Y_{f_0}(A_f - B_f B_f^\top X_{f_0}).$$

□

Moreover, the following result provides an upper norm-bound on the first approximation that still permits having $X_{f_1} \leq X_{f_0}$.

Kenney convergence theorems III

Theorem (Kenney, Laub, and Wette; 1990)

Let $X_{f_{N_{NK}}}$ *satisfy* $CARE_f$, $\hat{A}_f = A_f - B_f B_f^\top X_{f_{N_{NK}}}$ *is stable, and assume that* $\left\| X_{f_{N_{NK}}} - X_{f_0} \right\|_2 < r$, *where*

$$r = \frac{1}{3\|B_f\|_2^2 \left\| \hat{\Omega}_f^{-1} \right\|_2},$$

and the residual R_f, *defined as*

$$R_f \triangleq A_f^\top X_{f_0} + X_{f_0} A_f - X_{f_0} B_f B_f^\top X_{f_0} + Q,$$

satisfies

$$4\|B_f\|_2^2 \left\| \hat{\Omega}_f^{-1} \right\|_2^2 \|R_f\|_2 < 1;$$

Kenney convergence theorems IV

Theorem (Kenney, Laub, and Wette; 1990 – cont'd)

then

$$\left\| X_{f_{N_{NK}}} - X_{f_0} \right\|_2 \leq \frac{2\left\| \hat{\Omega}_f^{-1} \right\|_2 \|R_f\|_2}{1 + \sqrt{1 - 4\|B_f\|_2^2 \left\| \hat{\Omega}_f^{-1} \right\|_2 \|R_f\|_2}} \leq 2\left\| \hat{\Omega}_f^{-1} \right\|_2 \|R_f\|_2 \,.$$

□

Furthermore, the following result introduces another condition to having $X_{f_1} \leq X_{f_0}$.

Kenney convergence theorems V

Theorem (Kenney, Laub, and Wette; 1990)

Let $X_{f_{N_{NK}}}$ *satisfy* $CARE_f$, $\hat{A}_f = A_f - B_f B_f^\top X_{f_{N_{NK}}}$ *is stable, and assume that*

$$\left\| X_{f_{N_{NK}}} - X_{f_0} \right\|_2 < r \triangleq \frac{1}{3\|B_f\|_2^2 \left\| \hat{\Omega}_f^{-1} \right\|_2} ; \tag{4}$$

then

$$\left\| X_{f_{N_{NK}}} - X_{f_1} \right\|_2 \leq \left\| X_{f_{N_{NK}}} - X_{f_0} \right\|_2 , \tag{5}$$

where X_{f_1} *satisfies the Newton-Kleinman iteration for* $k = 1$, *i.e.,*

$$\hat{A}_{f_1}^\top X_{f_1} + X_{f_1} \hat{A}_{f_1} + K_{f_1}^\top R K_{f_1} + Q = 0 ;$$

Kenney convergence theorems VI

Theorem (Kenney, Laub, and Wette; 1990 – cont'd)

previously, (5) is an equality only when $X_{f_0} = X_{f_{N_{NK}}}$, *where* $\left\|\hat{\Omega}_f^{-1}\right\|_2$ *is the operator-norm of the inverse closed-loop Lyapunov-operator, i.e.,*

$$\hat{\Omega}_f(Y_f) \triangleq (A_f - B_f B_f^\top X_{f_{N_{NK}}})^\top Y_f + Y_f(A_f - B_f B_f^\top X_{f_{N_{NK}}}), \tag{6}$$

computed upon the solution to

$$(A_f - B_f B_f^\top X_{f_{N_{NK}}})^\top Y_f + Y_f(A_f - B_f B_f^\top X_{f_{N_{NK}}}) + I_n = 0, \tag{7}$$

by setting $\left\|\hat{\Omega}_f^{-1}\right\|_2 = \left\|\hat{\Omega}_f^{-1}(-I_n)\right\|_2$, *where the operator-norm is defined as*

$$\left\|\hat{\Omega}_f\right\|_2 = \max_{Y_f \in \mathbb{R}^{n\times n}_{Y_f \neq 0}} \frac{\left\|\hat{\Omega}_f(Y_f)\right\|_2}{\|Y_f\|_2}.$$

□

Remarks on Newton-Kleinman method I

Remark

Previously, it was considered that $r > 0$ is the supremum over all radii of the balls $\mathscr{B}_r\left(X_{f_{N_{NK}}}\right)$ centered at $X_{f_{N_{NK}}}$, such that, if $X_{f_0} \in \mathscr{B}_r(X_{f_{N_{NK}}})$, then

$$(A_f - B_f B_f^\top X_{f_0})^\top X_{f_1} + X_{f_1}(A_f - B_f B_f^\top X_{f_0}) =$$
$$-X_{f_0} B_f B_f^\top X_{f_0} - Q + (X_{f_{N_{NK}}} - X_{f_0}) B_f B_f^\top (X_{f_{N_{NK}}} - X_{f_0})$$

is uniquely solvable for X_{f_1} and $\left\| X_{f_{N_{NK}}} - X_{f_1} \right\|_2 < \left\| X_{f_{N_{NK}}} - X_{f_0} \right\|_2$. $\square$

Remarks on Newton-Kleinman method II

Remark

Note also that the control weighting matrix was taken unity before, $R_n = I_n$, since making the change of control variable as $\tilde{u}(t) = R_n^{1/2} u(t)$, the integral quadratic cost in the Linear Quadratic Regulator (LQR) problem indeed becomes

$$J[x_0, u(t)] = \int_0^{+\infty} \left[x^\top(t) Q_n x(t) + \|\tilde{u}(t)\|_2^2 \right] dt \,, \tag{8}$$

but the penalty matrix R_n vanish in the analysis of the post-fault CARE, which however does not affect the controllability of the system pair (A_f, B_f). $\square$

Newton-Raphson Method I

For the sake of its history from functional spaces, respect must be paid to the following result, as well.

Newton-Raphson Method II

Theorem (Kantorovich and Akilov; 1982)

Consider the function $f : \mathbb{R}^n \to \mathbb{R}^n$ *and let the following assumptions hold.*

There are β, η, $\tilde{r}$, *and* γ, *with* $\beta\eta\gamma \leq 1/2$, *and* $x_0 \in \mathbb{R}^n$, *such that:*

(i) f *is differentiable at* x_0, *respectively*

$$\left\| f'(x_0)^{-1} \right\|_2 \leq \beta \quad \textit{and} \quad \left\| f'(x_0)^{-1} f(x_0) \right\|_2 \leq \gamma \,;$$

(ii) f' *is Lipschitz continuous with Lipschitz-constant* γ *in a ball of radius* $\tilde{r} \geq r_-$ *about* x_0, *where*

$$r_- = \frac{1 - \sqrt{1 - 2\beta\eta\gamma}}{\beta\gamma} \,,$$

and define the ball $\mathscr{B}_0$ *of radius* r_- *about* x_0 *as*

$$\mathscr{B}_0 = \{x \mid \|x - x_0\|_2 \leq r_-\} \,;$$

then there is a unique root x^* *of* f *in* $\mathscr{B}_0$ *for all* k *in the Newton iteration.*

Newton-Raphson Method III

Theorem (Kantorovich and Akilov; 1982 – cont'd)

Solution x^* *is the unique root of* f *in the ball of radius*

$$\min\left\{\tilde{r}, \frac{1+\sqrt{1-2\beta\eta\gamma}}{\beta\gamma}\right\}$$

about x_0 *and the errors satisfy*

$$\|e_k\|_2 \leq \frac{(2\beta\eta\gamma)^{2n}}{2^n\beta\gamma}\,.$$

□

Newton-Kleinman algorithm I

This method established itself as an appropriate means to solve a CARE in matrix spaces as an extension of the Newton method in function spaces.

In essence, the main computational task at each iteration is to solve a linear algebraic matrix equation of Lyapunov-type and the stabilizing solution to the classical continuous algebraic Riccati equation is obtained as the limit of the Lyapunov equations solutions sequence.

This approach is based on the following result.

Newton-Kleinman algorithm II

Theorem (Kleinman; 1968)

Let X_{f_k} *be the unique positive-semidefinite solution of*

$$\hat{A}_{f_k}^\top X_{f_k} + X_{f_k}\hat{A}_{f_k} + K_{f_k}^\top R_n K_{f_k} + Q_n = 0\,, \tag{9}$$

where, for $k = 1, 2, \ldots, N_{NK}$*, recursively*

$$\begin{aligned} K_{f_k} &= R_n^{-1} B_f^\top X_{f_{k-1}}\,, \\ \hat{A}_{f_k} &= A_f - B_f K_{f_k}\,, \end{aligned} \tag{10a}$$

and K_{f_0} *chosen such that the initial closed-loop matrix is stable, that is* $\hat{A}_{f_0} = A_f - B_f K_{f_0}$ *has eigenvalues with negative real-parts.*

Newton-Kleinman algorithm III

Theorem (Kleinman; 1968 – cont'd)

Then

$$\lim_{k \to +\infty} X_{f_k} = X_{f_{N_{NK}}} \,, \tag{11a}$$

with

$$0 \leq X_{f_{N_{NK}}} \leq \ldots \leq X_{f_{k+1}} \leq X_{f_k} \leq \ldots \leq X_{f_1} \,. \tag{11b}$$

□

Newton-Kleinman algorithm IV

From the previous theorem, it follows that, given any initial stabilizing matrix K_{f_0}, the solution to the continuous algebraic Lyapunov equation

$$\hat{A}_{f_{N_{NK}}}^{\top} X_{f_{N_{NK}}} + X_{f_{N_{NK}}} \hat{A}_{f_{N_{NK}}} + K_{f_{N_{NK}}}^{\top} R_n K_{f_{N_{NK}}} + Q_n = 0\,,$$

for k large enough, converges to that of the continuous algebraic Riccati equation

$$A_f^{\top} X_{f_{N_{NK}}} + X_{f_{N_{NK}}} A_f - K_{f_{N_{NK}}}^{\top} R_n K_{f_{N_{NK}}} + Q_n = 0\,; \tag{12}$$

removing index N_{NK}, remark that (12) and (13), respectively

$$CARE_f: \quad A_f^{\top} X_f + X_f A_f - X_f B_f R_n^{-1} B_f^{\top} X_f + Q_n = 0\,, \tag{13}$$

are identical.

Newton-Kleinman algorithm V

This algorithm provides to have Newton-like quadratic rate, i.e.,

$$\left\| X_{f_{N_{NK}}} - X_{f_{k+1}} \right\|_2 \leq c \left\| X_{f_{N_{NK}}} - X_{f_k} \right\|_2^2,$$

as it approaches convergence, where c is a constant independent of the iteration index k, and $\|.\|_2$ stands for the matrix 2-norm.

In general, $X_{f_0} \geq X_{f_1}$ is not true.

This far, the computational load accumulated by 8-10 iterations performed by the Newton-Kleinman method is equivalent to that provided by the Schur Vector Method used to solving the post-fault CARE.

Outline

1. Iterative Methods for Fault Accommodation
2. Initial Step Problem for the Newton-Kleinman Method
   - Kleinman Initialization Method
   - Bass-Armstrong Initialization Method
   - Numerical Solution to the Initial Step Problem
3. B747 Application Example

Kleinman initialization method I

If the pair (A_f, B_f) remains completely controllable, then $u_f(t) = -K_f x(t)$ is a stabilizing control law with K_f given by the following result.

Theorem (Kleinman; 1970)

If the system $\dot{x}(t) = A_f x(t) + B_f u_f(t)$ is completely controllable, then $u_f(t) = -K_f x(t)$ is a stabilizing control law, with

$$K_f = B_f^\top W_f^{-1}(T)\,, \;\; \textit{for} \tag{14a}$$

$$W_f(T) = \int_0^T e^{-A_f t} B_f B_f^\top e^{-A_f^\top t}\, dt\,, \;\; \textit{with} \;\; T = \textit{arbitrary}\,, \tag{14b}$$

and

$$V_f[x(t)] = x^\top(t) W_f^{-1}(T) x(t)$$

is a suitable Lyapunov function for the closed-loop system. □

Kleinman initialization method II

In any event, the previous theorem must be seen as the concretization of two classical results from the Lyapunov theory on the stability of linear systems.

Unfortunately, choosing the value of T in (14b) depends on the magnitude of the fault, in the eventuality that such an estimation is provided by the FDIE module; it requires a tradeoff to be performed between the accuracy of the integration result and the time available for obtaining the initial stabilizing feedback before the situation becomes critical.

Note however that $W_f(T)$ in (14b) represents the controllability Gramian of the system pair (A_f, B_f), i.e., the solution to the continuous algebraic Lyapunov equation

$$(-A_f)W_f + W_f(-A_f)^\top + B_f B_f^\top = 0\,, \tag{15}$$

when taking the integration limit $T \to +\infty$; still, it is unsure whether such a stabilizing solution always exists if the system matrix A_f has null eigenvalues after the fault.

This fact is to be seen with the help of the Kronecker (tensor) product $\otimes$ of two matrices; also, the vec operator of a matrix A is introduced to create a column vector $\mathrm{vec}\,(A)$ that puts the columns of A from left to right one in top of another.

Kleinman initialization method III

Hence, given the matrices $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{m\times m}$, the Kronecker product $A \otimes B$ produces another matrix as follows

$$A \otimes B = \begin{bmatrix} a_{11}B & \dots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \dots & a_{nn}B \end{bmatrix};$$

moreover, if matrix A is the identity matrix, the result of $I_n \otimes B$ is a block-diagonal matrix with B repeated n times along the diagonal, whose resulting matrix spectrum contains n times the eigenvalues of B.

Furthermore, using the property of the Kronecker product which states that, if A, B, C and X are matrices of appropriate dimensions such that $C = AXB$, then $\mathrm{vec}\,(C) = (B^\top \otimes A)\mathrm{vec}\,(X)$, that translates, in the case of (15), into

$$[(I_n \otimes A_f) + (A_f \otimes I_n)]\,\mathrm{vec}\,(W_f) = \mathrm{vec}\left(B_f B_f^\top\right), \tag{16}$$

and the system of equations (16) has a solution for W_f if $[(I_n \otimes A_f) + (A_f \otimes I_n)]$ is nonsingular, therefore imposing the obvious condition on the eigenvalues of A_f.

Bass-Armstrong initialization method I

An extension to (15), and implicitly on (16), firstly operates a shift on the diagonal elements of the post-fault system matrix A_f; in this case, W_f is found based on the following result.

Theorem (Armstrong; 1975)

Let (A_f, B_f) *be controllable.*

Then $K_f = B_f^\top W_f^{-1}$ *stabilizes the system* $\dot{x}(t) = A_f x(t) + B_f u_f(t)$, *where* $W_f^\top = W_f > 0$ *satisfies*

$$[-(A_f + \beta_f I_n)]\, W_f + W_f\, [-(A_f + \beta_f I_n)]^\top + 2 B_f B_f^\top = 0\,, \tag{17}$$

with $\beta_f > \|A_f\|_2$. $\square$

Bass-Armstrong initialization method II

The choice $\beta_f > \|A_f\|_2$ eventually counteracts the instability of A_f, ensuring that matrices $(A_f+\beta_f I_n)$ and $[-(A_f+\beta_f I_n)]$ share no common eigenvalues and that the Lyapunov operator $\Omega_f(W_f) = A_f W_f + W_f A_f^\top$ becomes nonsingular and thus invertible, that is (17) has a solution. In that case, β_f gives the "$-\beta_f$ degree-of-stability", i.e., $\Lambda(A_f - B_f K_f) \in \mathbb{C}^-_{-\beta_f}$.

To see this, consider the partition of the complex-plane $\mathbb{C}$ determined by the imaginary axis $\jmath\omega$, that is

$$\mathbb{C} = \mathbb{C}^- \oplus \mathbb{C}^+ \,,\ \mathbb{C}^- \cap \mathbb{C}^+ = \emptyset \,,$$

with

$$\mathbb{C}^- = \{s \in \mathbb{C} \mid \mathfrak{Re}\,(s) < 0\} \,,\ \mathbb{C}^+ = \{s \in \mathbb{C} \mid \mathfrak{Re}\,(s) \geq 0\} \,,$$

where $s = \sigma + \jmath\omega$ is a complex variable with the real and imaginary parts $\mathfrak{Re}\,(s) = \sigma$ respectively $\mathfrak{Im}\,(s) = \omega$, and $\emptyset$ denotes the empty set.

Bass-Armstrong initialization method III

Moreover, in the continuous case, if K_f is a stabilizing matrix for the pair $(A_f+\beta_f I_n, B_f)$, with $\beta_f > 0$, then the same control matrix achieves $\Lambda(A_f - B_f K_f) \in \mathbb{C}^-_{-\beta_f}$, for a partition of the complex-plane according to desired stability and dynamic properties of the closed-loop, that is w.r.t. the axis defined by $-\jmath\beta_f\omega$, i.e.,

$$\mathbb{C} = \mathbb{C}^-_{-\beta_f} \oplus \mathbb{C}^+_{-\beta_f}\,,\ \mathbb{C}^-_{-\beta_f} \cap \mathbb{C}^+_{-\beta_f} = \emptyset\,,$$

for

$$\mathbb{C}^-_{-\beta_f} = \{s \in \mathbb{C} \mid \mathfrak{Re}\,(s) < -\beta_f\}\,,\ \mathbb{C}^+_{-\beta_f} = \{s \in \mathbb{C} \mid \mathfrak{Re}\,(s) \geq -\beta_f\}\,.$$

Furthermore, for completeness of the analysis, look for the arguments on the validity of a stabilizing initial guess if replacing (17) with

$$[-(A_f + \beta_f I_n)]\, W_f + W_f [-(A_f + \beta_f I_n)]^\top + \eta_f B_f B_f^\top = 0\,,$$

for $0 < \eta_f \leq 2$, but the upper limit 2 provides a minimum norm for the control matrix K_f.

Bass-Armstrong initialization method IV

However, the above process is seen as inefficient somehow because it does the shift β_f on all eigenvalues, moving also the eigenvalues that might be in the "good" region of $\mathbb{C}^-$; that is why partial stabilizing algorithms are recommended sometimes, which means procedures that conduct only the unstable eigenvalues to target regions with certain dynamic properties.

This far, the restrictive assumption of complete controllability of (A_f, B_f) required by the previous two theorems has been used. However, similar results relax this condition to stabilizability only; in that case, W_f^{-1} is replaced by the pseudo-inverse matrix $W_f^{\dagger}$.

Numerical solution to the initial step problem I

Efficient algorithms to implementing the solution of the initial step problem basically demand for rapid and numerically stable solvers of Lyapunov equations.

Consider therefore the Lyapunov equation as required by the Bass-Armstrong approach in (17) and denote $\bar{A}_f = [-(A_f + \beta_f I_n)]$ to obtain

$$\bar{A}_f W_f + W_f \bar{A}_f^\top + 2 B_f B_f^\top = 0\,. \tag{18}$$

The Bartels-Stewart algorithm is based on the RSF of $\bar{A}_f$ and it stands for the most efficient implementation to solving (18); the reduction of $\bar{A}_f$ to upper-Hessenberg form adds no improvement to the technique.

Numerical solution to the initial step problem II

In this respect, transform $\bar{A}_f$ to its Real Schur Form by applying an orthogonal transform U_f such that

$$U_f^\top \bar{A}_f U_f U_f^\top W_f U_f + U_f^\top W_f U_f U_f^\top \bar{A}_f^\top U_f + 2 U_f^\top B_f B_f^\top U_f = 0\,;$$

then, denote the transformed matrices by $\bar{A}'_f = U_f^\top \bar{A}_f U_f$, $\bar{W}'_f = U_f^\top W_f U_f$, respectively $\bar{B}'_f = U_f^\top B_f$, and construct $\bar{C}'_f = 2\bar{B}'_f \bar{B}'^\top_f$.

Hence, partition the above matrices accordingly to obtain the following block-matrix relation

$$0 = \begin{bmatrix} \bar{A}'_{f_{11}} & \bar{A}'_{f_{12}} \\ O_{d,n-d} & \bar{A}'_{f_{22}} \end{bmatrix} \begin{bmatrix} \bar{W}'_{f_{11}} & \bar{W}'^\top_{f_{21}} \\ \bar{W}'_{f_{21}} & \bar{W}'_{f_{22}} \end{bmatrix} + \begin{bmatrix} \bar{W}'_{f_{11}} & \bar{W}'^\top_{f_{21}} \\ \bar{W}'_{f_{21}} & \bar{W}'_{f_{22}} \end{bmatrix} \begin{bmatrix} \bar{A}'^\top_{f_{11}} & O_{d,n-d} \\ \bar{A}'^\top_{f_{12}} & \bar{A}'^\top_{f_{22}} \end{bmatrix} \tag{19}$$

$$+ \begin{bmatrix} \bar{C}'_{f_{11}} & \bar{C}'_{f_{12}} \\ \bar{C}'_{f_{21}} & \bar{C}'_{f_{22}} \end{bmatrix}, \tag{20}$$

where $d = \dim(\bar{A}'_{f_{22}})$.

Numerical solution to the initial step problem III

Furthermore, write the block-matrix equation corresponding to the bottom-right corner in (19), that is

$$\bar{A}'_{f_{22}}\bar{W}'_{f_{22}} + \bar{W}'_{f_{22}}\bar{A}'^{\top}_{f_{22}} = -\bar{C}'_{f_{22}}\,, \tag{21}$$

and consider that $\bar{A}'_{f_{22}}$ is associated with a real eigenvalue, in which case

$$\bar{W}'_{f_{22}} = -\frac{1}{2}\frac{\bar{C}'_{f_{22}}}{\bar{A}'_{f_{22}}}$$

is obtained; if $\bar{A}'_{f_{22}}$ is associated with a pair of complex-conjugate eigenvalues, then the block-matrices turn to second order and (21) is replaced by a system of equations of order three only due to symmetry of $\bar{W}'_{f_{22}}$ and $\bar{C}'_{f_{22}}$. The process continues in the same manner to the top-left corner of (19), but note that some Sylvester equations must also be solved at some point.

This far, the total amount of work performed by the Bartels-Stewart algorithm is averaged at $14\,n^3$.

Outline

1. Iterative Methods for Fault Accommodation
2. Initial Step Problem for the Newton-Kleinman Method
3. B747 Application Example
   - Iterative Fault Accommodation - Newton-Kleinman Approach
   - Faulty System - Unstable Case

Iterative fault accommodation - Newton-Kleinman approach I

Even in the faulty case, using the nominal control matrix of the nominal CARE, the closed-loop system $\hat{M}_f$, with the closed-loop matrix $\hat{A}_f = A_f - B_f K_n$ as

$$\hat{A}_f = \begin{bmatrix} -0.4525 & 679.4511 \\ -0.0119 & -1.4921 \end{bmatrix},$$

remains stable, with the spectrum

$$\Lambda(\hat{A}_f) = \{-0.9723 \pm 2.8004\jmath\}.$$

Still, it is desirable to improve the system closed-loop behavior based on the solutions provided at every iteration step by the Newton-Kleinman method.

In this respect, K_n may be chosen as the stabilizing initial gain of the algorithm. Convergence is reached when there are no further variations in the elements of the feedback control matrix.

Iterative fault accommodation - Newton-Kleinman approach II

Hence, taking (22), precisely

$$K_n = \begin{bmatrix} -0.0954 & -9.5582 \end{bmatrix}, \tag{22}$$

as the initial gain of the Newton-Kleinman approach, it converges to the solution of the post-fault Continuous Algebraic Riccati Equation, respectively

$$CARE_f: \quad A_f^\top X_f + X_f A_f - X_f B_f R_n^{-1} B_f^\top X_f + Q_n = 0\,,$$

that is

$$X_f^{acc} = \begin{bmatrix} 0.0040 & 0.6387 \\ 0.6387 & 221.7001 \end{bmatrix}, \tag{23}$$

which gives the accommodated state-feedback matrix

$$K_f^{acc} = \begin{bmatrix} -0.0812 & -26.8122 \end{bmatrix}. \tag{24}$$

Iterative fault accommodation - Newton-Kleinman approach III

Then, the accommodated closed-loop system $\hat{M}_f^{acc}$, with the closed-loop matrix $\hat{A}_f^{acc} = A_f - B_f K_f^{acc}$ as

$$\hat{A}_f^{acc} = \begin{bmatrix} -0.4270 & 648.4947 \\ -0.0103 & -3.4896 \end{bmatrix},$$

is stable, with the spectrum

$$\Lambda(\hat{A}_f^{acc}) = \{-1.9583 \pm 2.0813\jmath\}.$$

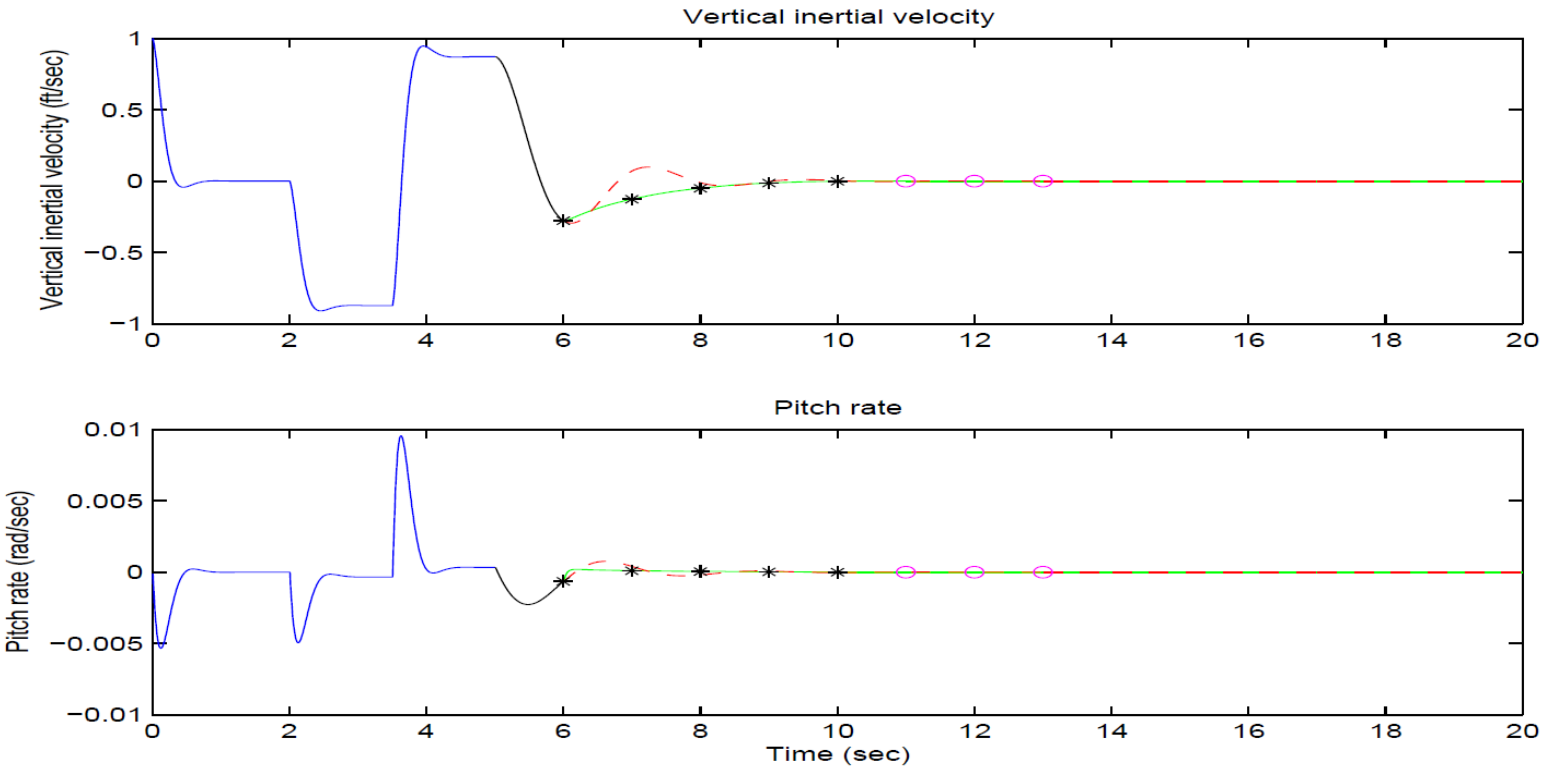
The following table shows the complete accommodation of the control matrix $K_f^{acc} = \begin{bmatrix} k_{f_1}^{acc} & k_{f_2}^{acc} \end{bmatrix}$ along the $N_{NK} = 8$ iterations.

k	1	2	3	4
$k_{f_1}^{acc}$	-0.1778	-0.1024	-0.0783	-0.0784
$k_{f_2}^{acc}$	-223.144	-117.248	-60.4003	-35.2454

k	5	6	7	8
$k_{f_1}^{acc}$	-0.0808	-0.0812	-0.0812	-0.0812
$k_{f_2}^{acc}$	-27.6509	-26.8222	-26.8122	-26.8122

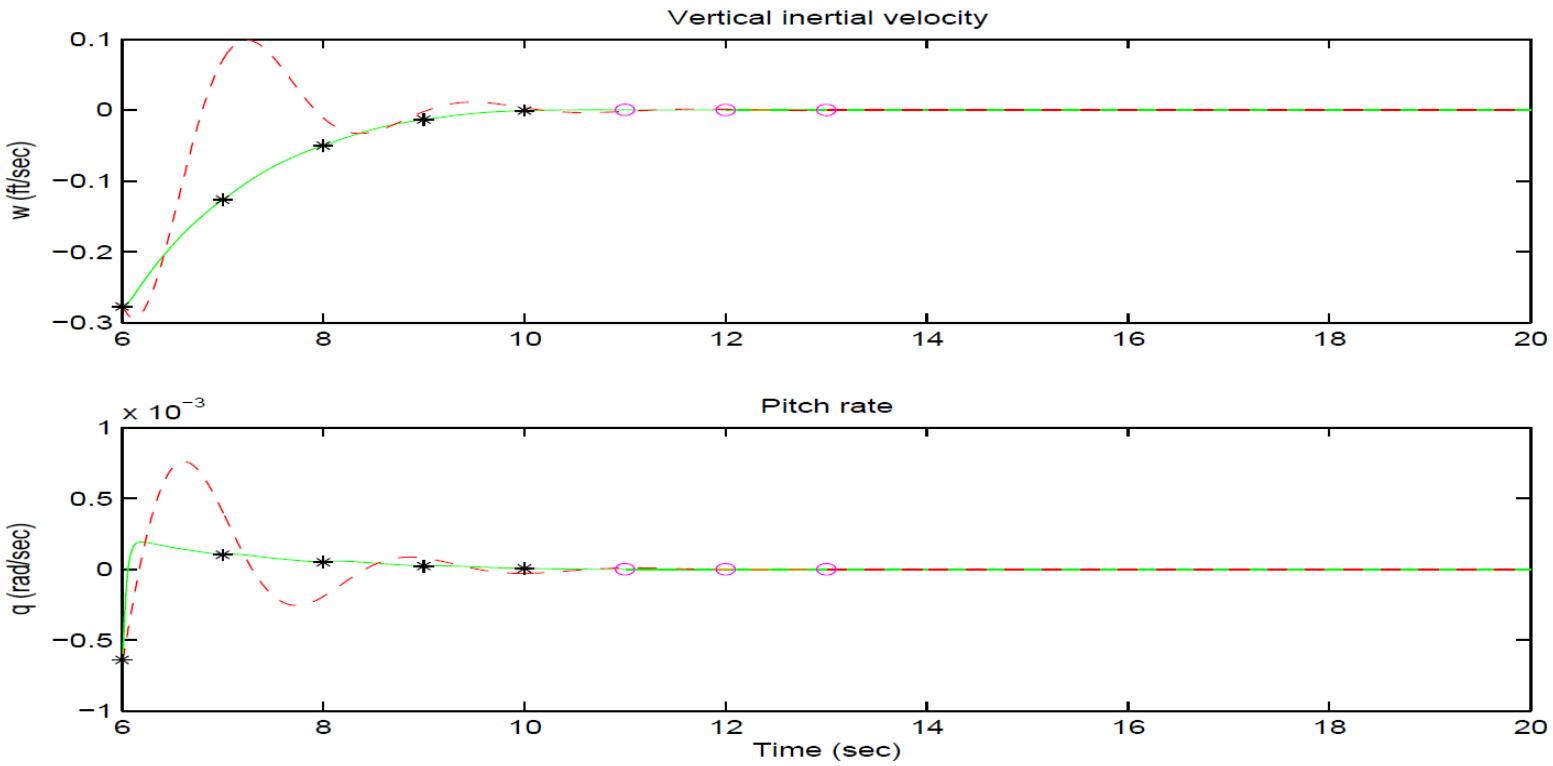
Iterative fault accommodation - Newton-Kleinman approach IV

The following figure displays the simulations of the closed-loop behavior during the post-fault period when applying the progressive and the classical control schemes (for simplicity, the samples $\bar{\Delta}_k = 1$ *sec* were considered). The fault occurred at $t_f = 4$ *sec*, the estimation of the faulty model was ready at $t_{FDIE} = 5$ *sec*, therefore there is one sample of uncertain evolution, until $t_{FDIE+\Delta} = 6$ *sec*.



NK-PAS vs. CAS for B747-SP - Stable Case (Continuous)

Iterative fault accommodation - Newton-Kleinman approach V



NK-PAS vs. CAS for B747-SP - Stable Case (Transient)

Iterative fault accommodation - Newton-Kleinman approach VI

Only the transient, after that instant, is highlighted in the previous figure. Remark therefore that progressive accommodation proceeds smoother than the classical approach. (The computation time for classical accommodation using the direct method based on the Schur Vector Method was assimilated with that needed by approximately 8 steps in the progressive accommodation based on Newton-Kleinman approach.)

Also, see in the following table the evolution of the accommodated matrix K_f^{acc} and the corresponding number of steps N_{NK} until convergence, for different choices of τ_f; hence, observe that N_{NK} becomes larger as τ_f increases.

τ_f	0.1	0.3	0.5	0.7	0.9
$k_{f_1}^{acc}$	-0.0954	-0.0946	-0.0933	-0.0907	-0.0812
$k_{f_2}^{acc}$	-9.5582	-10.9121	-12.9630	-16.6407	-26.8122
N_{NK}	4	5	6	7	8

Faulty system - Unstable case I

Consider again an actuator fault defined by the loss-of-effectiveness with $90\,\%$ in the elevator deflection. However, assume this time a structural fault modeled in the system matrix as $A_f = A_n + \Delta A_f \times I_2$, with $\Delta A_f = 773.9774 \times 10^{-2}$ computed upon the maximum singular value of A_n.

The faulty pair (A_f, B_f) is therefore given by

$$A_f = \begin{bmatrix} 7.4273 & 774.0000 \\ -0.0010 & 7.3116 \end{bmatrix} \text{ and } B_f = \begin{bmatrix} -1.7942 \\ -0.1158 \end{bmatrix}.$$

Iterative fault accommodation - Newton-Kleinman approach I

Note that, for the faulty pair (A_f, B_f) introduced above, the nominal control matrix K_n given in (22) does not ensure stability anymore.

The post-fault closed-loop system $\hat{M}_f$, with the closed-loop matrix $\hat{A}_f = A_f - B_f K_n$ as

$$\hat{A}_f = \begin{bmatrix} 7.2562 & 756.8511 \\ -0.0120 & 6.2051 \end{bmatrix},$$

has an unstable pair of poles in the spectrum

$$\Lambda(\hat{A}_f) = \{6.7306 \pm 2.9726\jmath\} .$$

That is, the initial stable step required by the progressive accommodation based on the Newton-Kleinman approach cannot be computed upon the nominal solution to CARE$_n$, as it works for faults which do not destabilize the system.

Iterative fault accommodation - Newton-Kleinman approach II

Use therefore the Bass-Armstrong approach with a scaled $\beta_f = 7.7407$ and obtain the initial control matrix

$$K_{f_0}^{acc} = \begin{bmatrix} -5.1073 & 181.8859 \end{bmatrix},$$

that provides the stable closed-loop system $\hat{M}_{f_0}^{acc}$, with the closed-loop matrix $\hat{A}_{f_0}^{acc} = A_f - B_f K_{f_0}^{acc}$ as

$$\hat{A}_{f_0}^{acc} = \begin{bmatrix} -15.1680 & -774.0000 \\ 0.0010 & -15.0523 \end{bmatrix},$$

with the stable spectrum

$$\Lambda(\hat{A}_{f_0}^{acc}) = \{-7.5714, -43.9329\};$$

observe that the eigenvalues lose their complex-conjugateness.

Iterative fault accommodation - Newton-Kleinman approach III

To decide whether the progressive accommodation achieved convergence, different criteria can be used, e.g., either tolerance / stagnation of the normalized residual norm or smallness / stagnation of the relative changes of control gains; in order to ensure 4-digits accuracy, a condition on the smallness of control gain differences between two iterations, i.e., $|k^i_{f_{j+1}} - k^i_{f_j}| \leq 10^{-5}$, for $j = 1 : N_{NK} - 1$ and $i = 1 : n$, was used.

The post-fault closed-loop still needed $N_{NK} = 8$ iterations for the accommodation of the post-fault closed-loop system. The evolution of control gains using Newton-Kleinman method is shown in the following table.

k	1	2	3	4
$k^{acc}_{f_1}$	-8.5772	-5.0054	-3.3148	-2.6294
$k^{acc}_{f_2}$	-439.2699	-309.8357	-248.4677	-223.4992

k	5	6	7	8
$k^{acc}_{f_1}$	-2.4623	-2.4510	-2.4510	-2.4510
$k^{acc}_{f_2}$	-217.3744	-216.9600	-216.9582	-216.9582

Questions? vs. Answers!

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